

NeSPReS0 v2_alpha multi-profile check

One POST per model. Same body for v1, A_CRPS, HeaveFast, and ops:
3 locations x 2 dates = 6 profiles, one-to-one (lat[i], lon[i], date[i]).

API http://127.0.0.1:5000

Locations

25.00 N,	83.00 W	east GoM, west of Florida
26.50 N,	85.50 W	north-central GoM
24.80 N,	86.50 W	south-central GoM

Dates 2016-12-15, 2025-06-15

Request body

```
{'lat': [25.0, 26.5, 24.8, 25.0, 26.5, 24.8], 'lon': [-83.0, -85.5, -86.5, -83.0, -85.5, -86.5],  
'date': ['2016-12-15', '2016-12-15', '2016-12-15', '2025-06-15', '2025-06-15', '2025-06-15']}
```

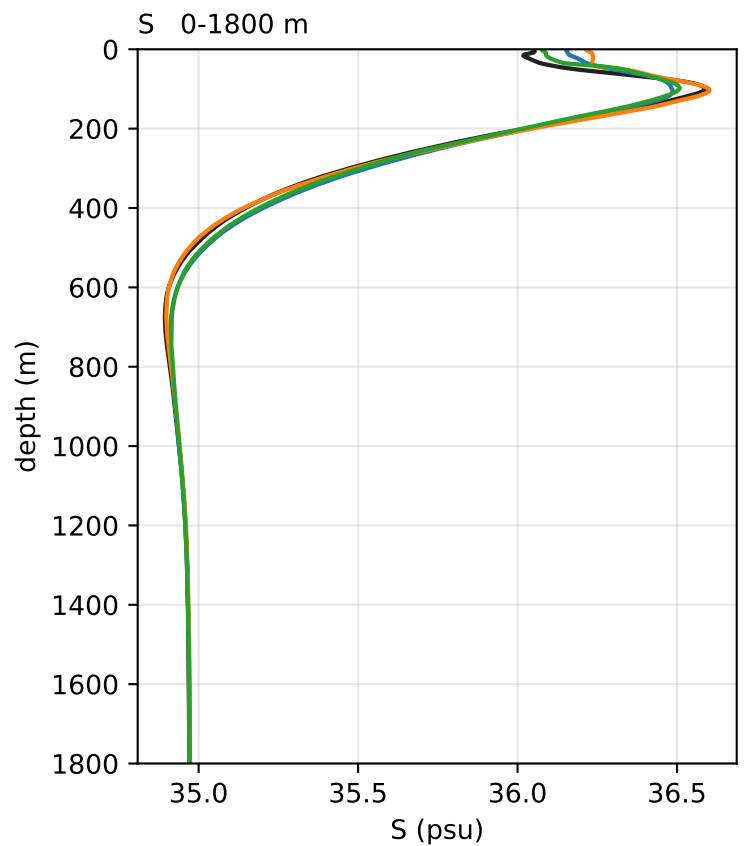
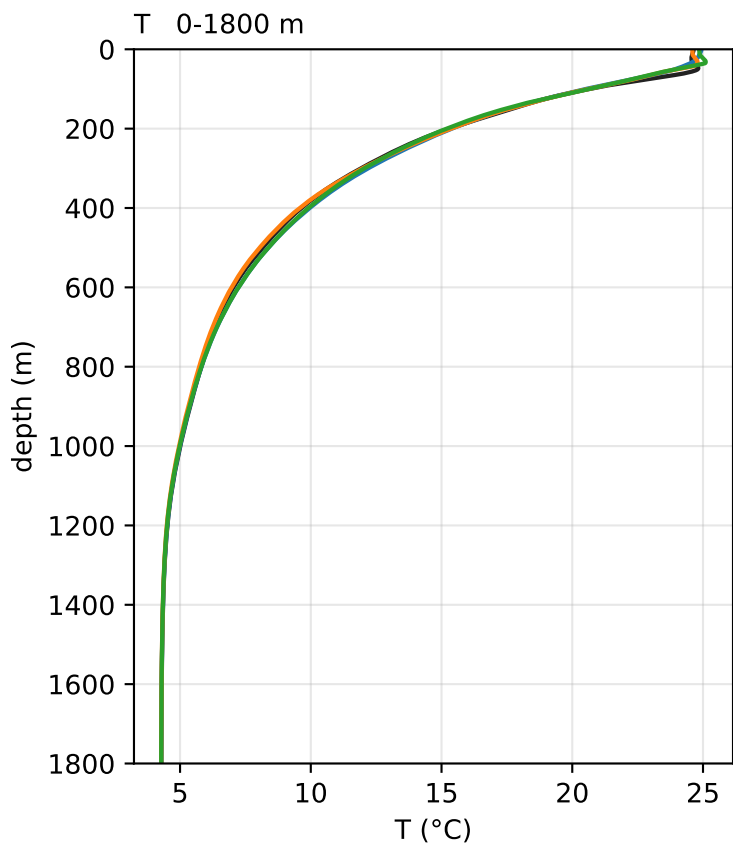
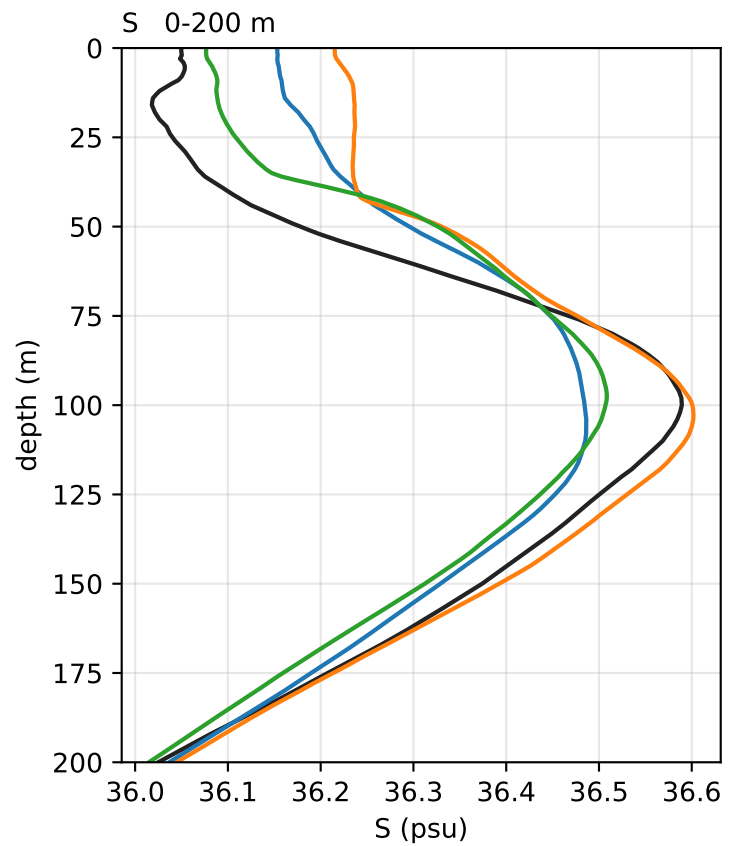
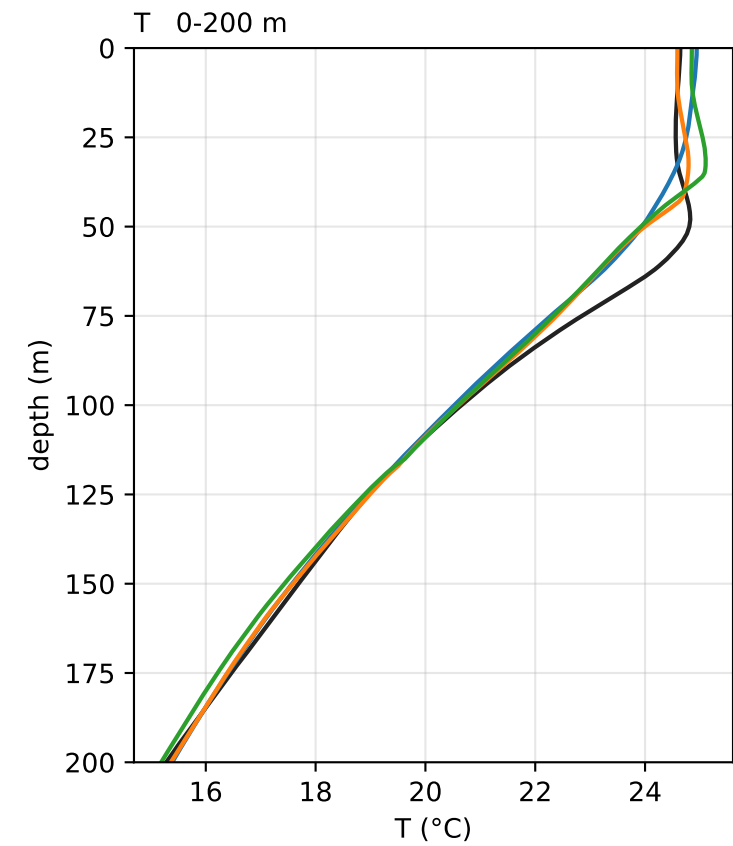
HTTP

v1	200	n=6	2.7s	model=none (SAT TorchScript)	SHA=unknown
A_CRPS	200	n=6	0.3s	model=A_CRPS	SHA=p5_A_CRPS_v2_s42_s2
HeaveFast	200	n=6	0.4s	model=HeaveFast	SHA=heave_fast_s42
ops	200	n=6	9.3s	model=ops	SHA=heave_ops_s42

Each later page is one location and date. Left column temperature,
right column salinity. Top row 0-200 m, bottom row 0-1800 m.
Depth increases downward. v1 is the Ozavala SAT cell (15 PCs).
A_CRPS / HeaveFast / ops are the frozen v2 DA cells.

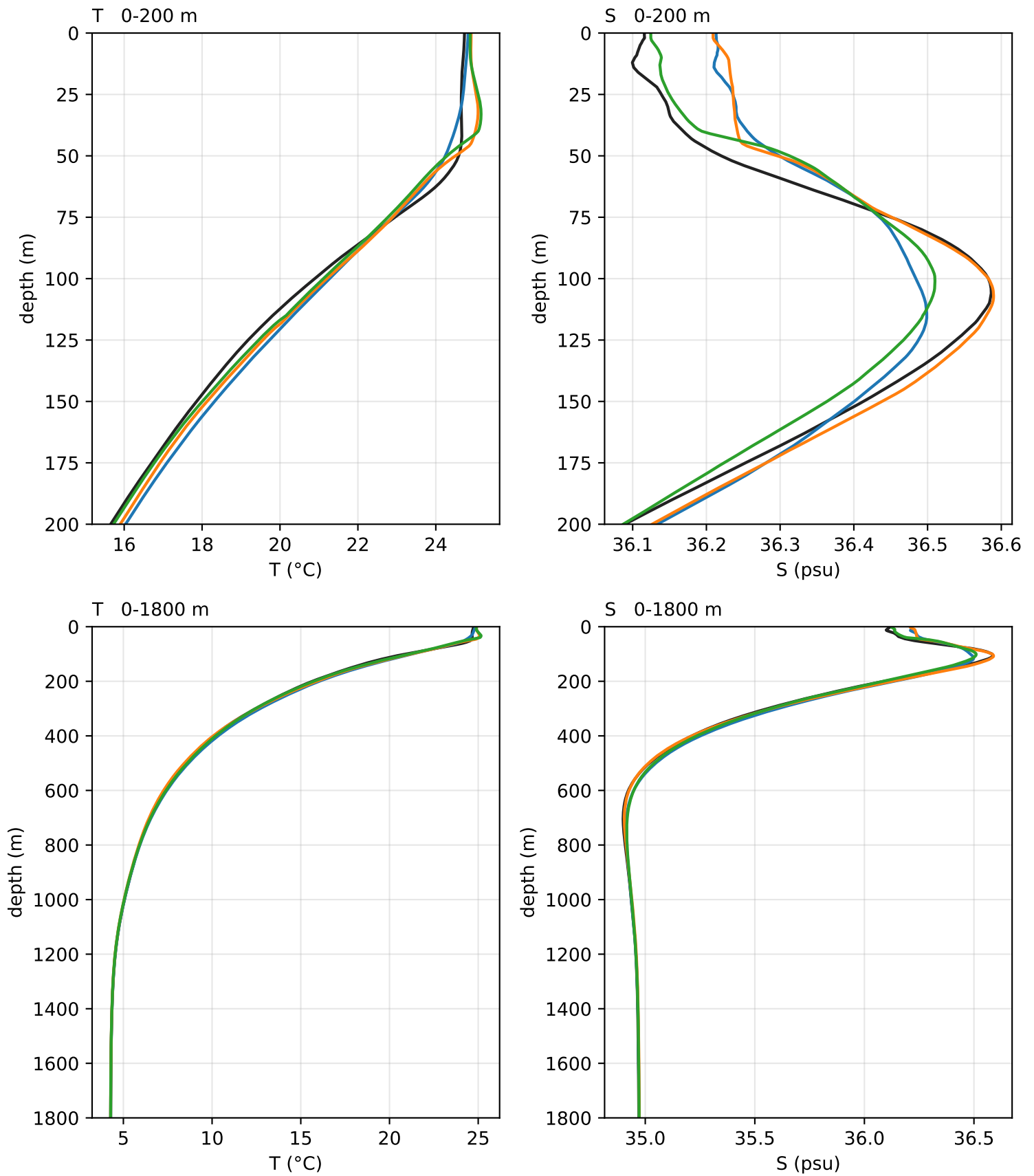
ops operators were computed from the live SAT archive at request time,
not from gom_cube.zarr. That is why 2025 is on this page at all.

25.00 N, 83.00 W 2016-12-15
east GoM, west of Florida



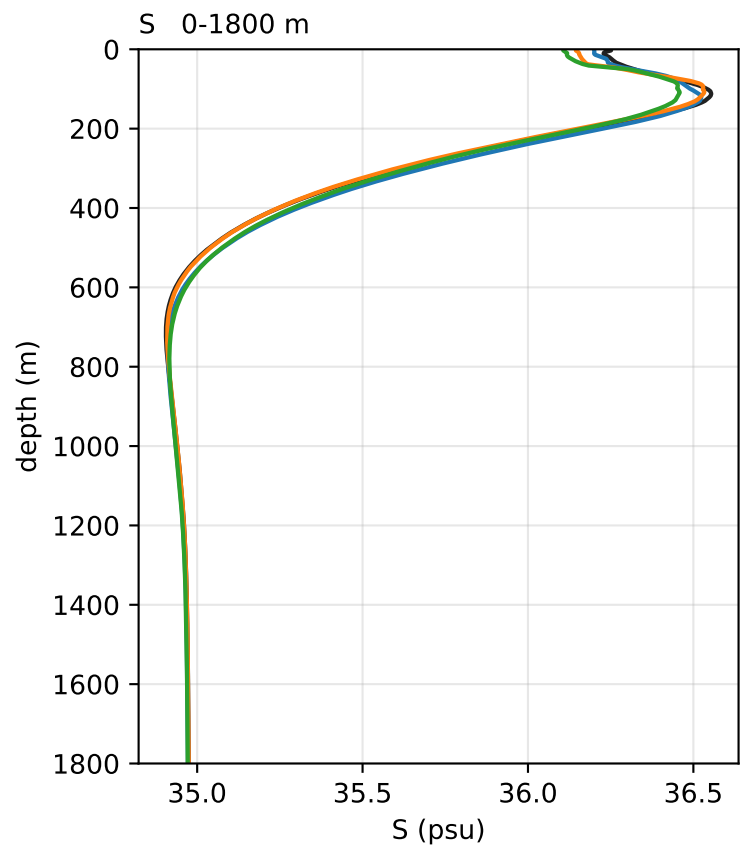
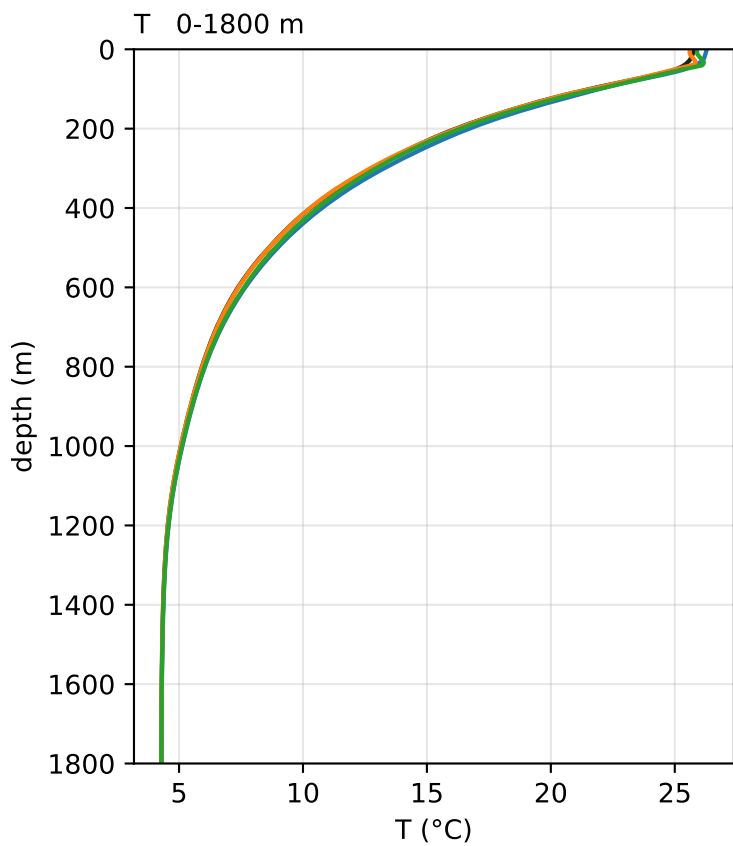
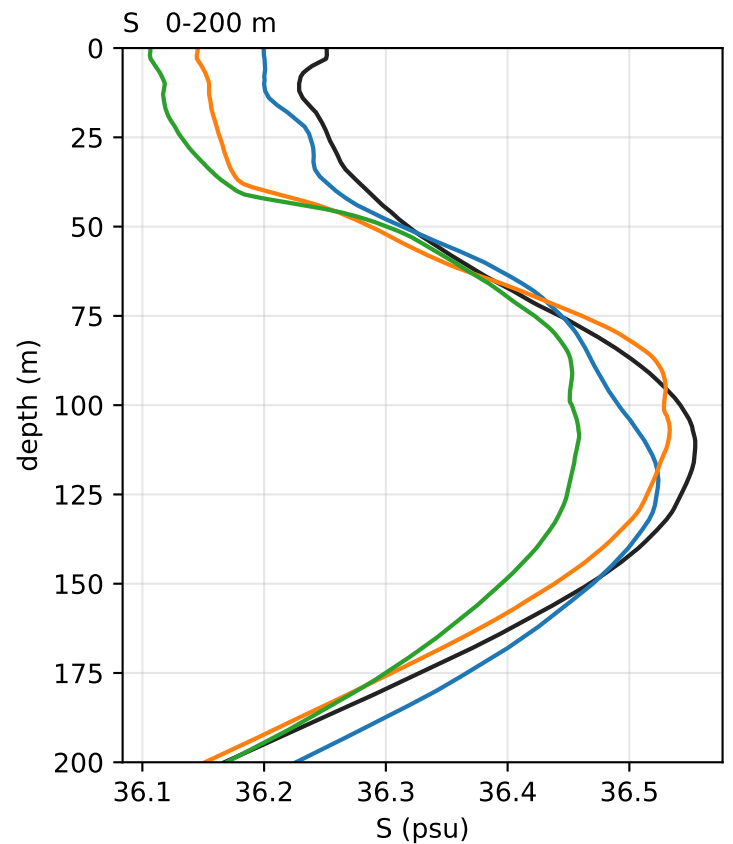
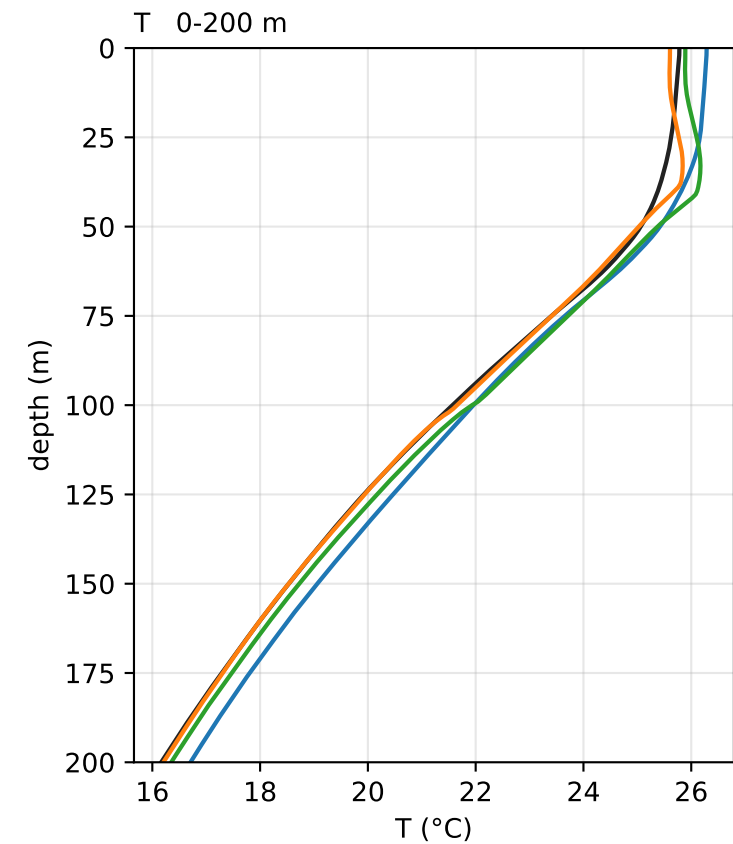
v1 A_CRPS HeaveFast ops

26.50 N, 85.50 W 2016-12-15
north-central GoM



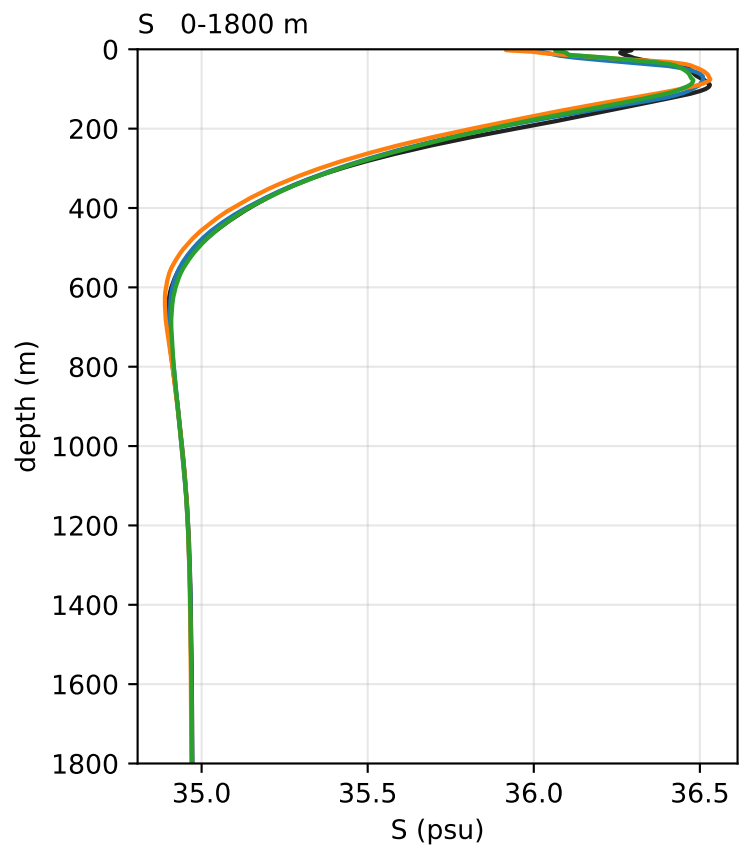
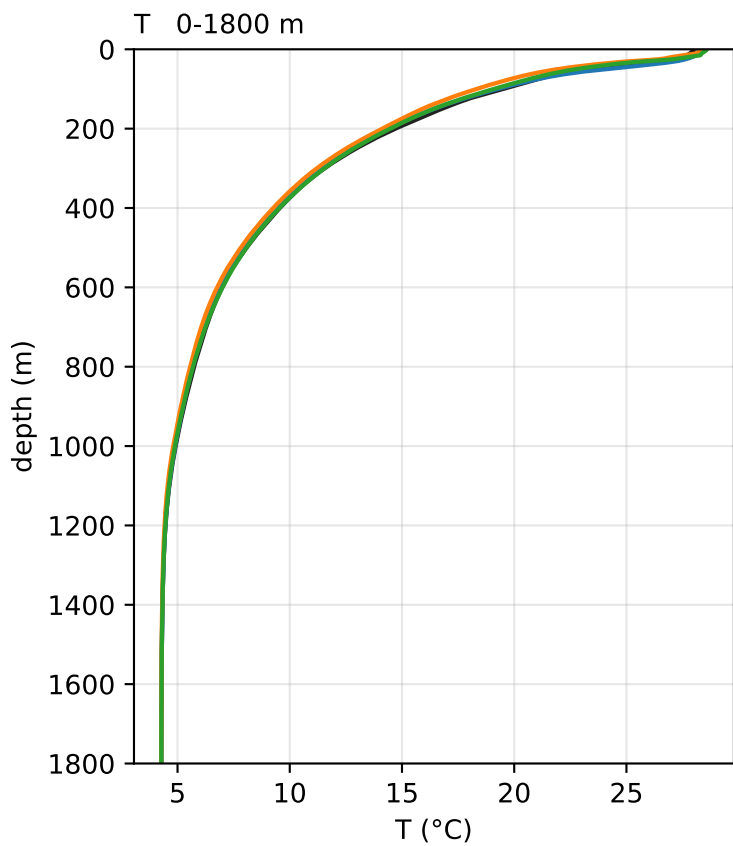
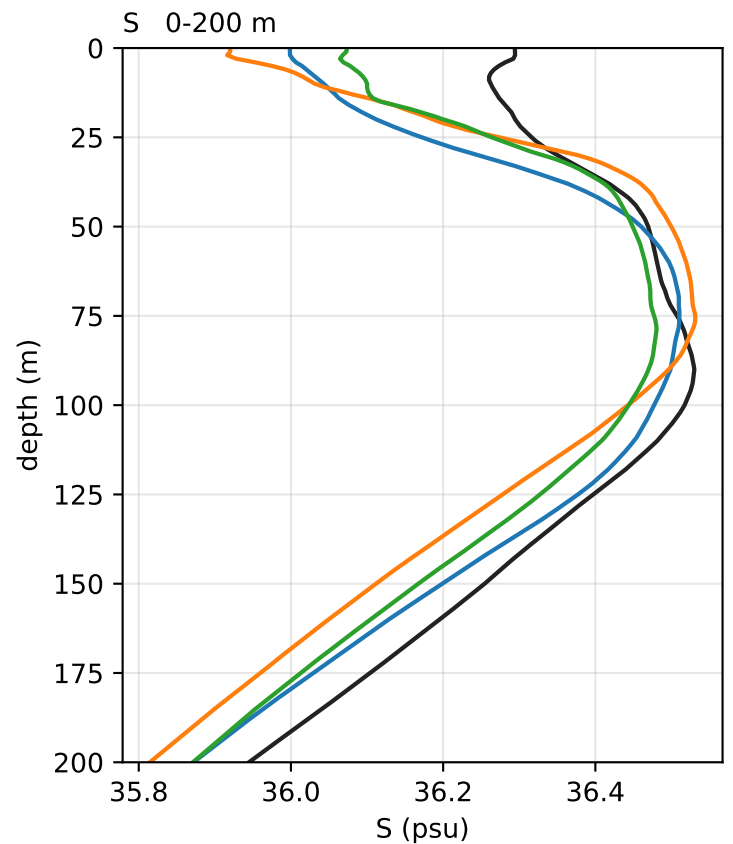
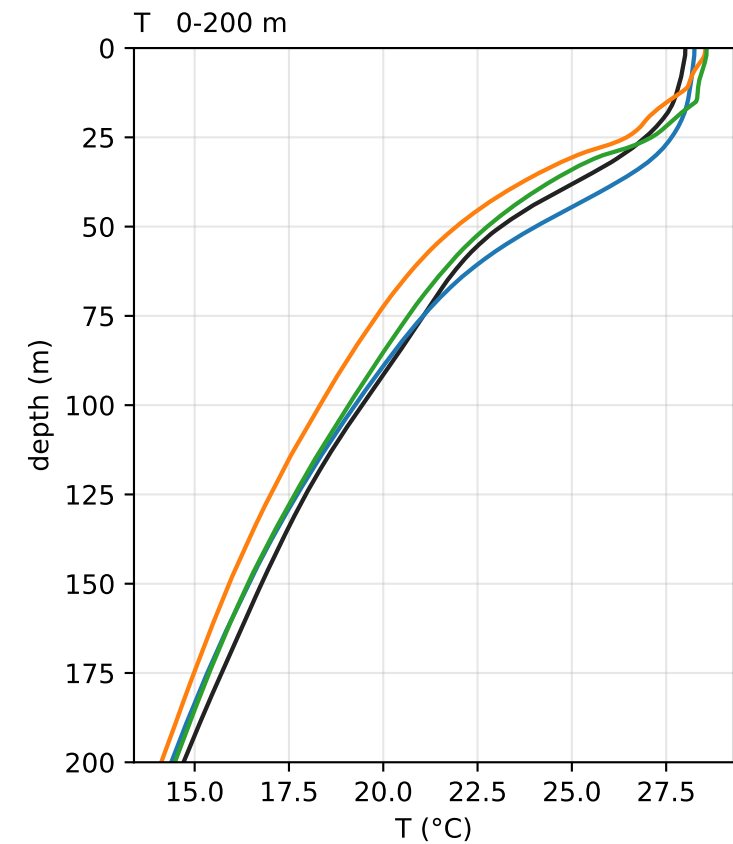
v1 A_CRPS HeaveFast ops

24.80 N, 86.50 W 2016-12-15
south-central GoM



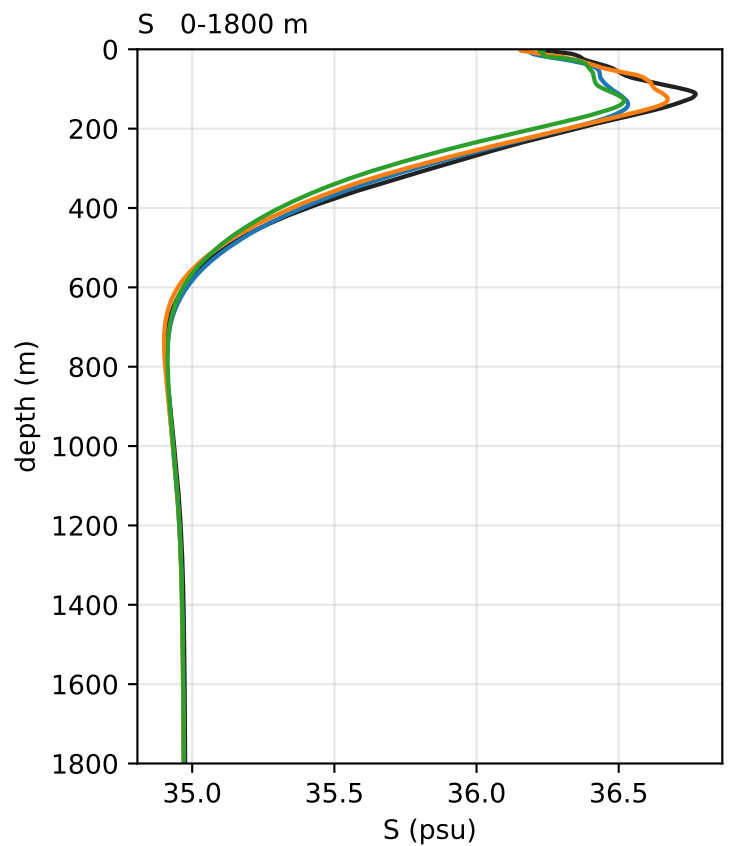
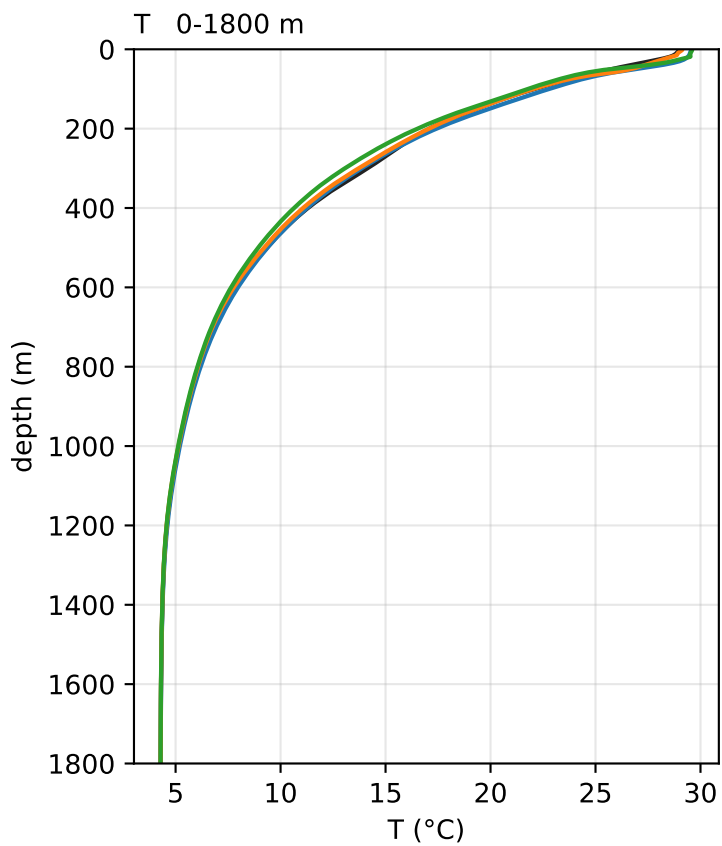
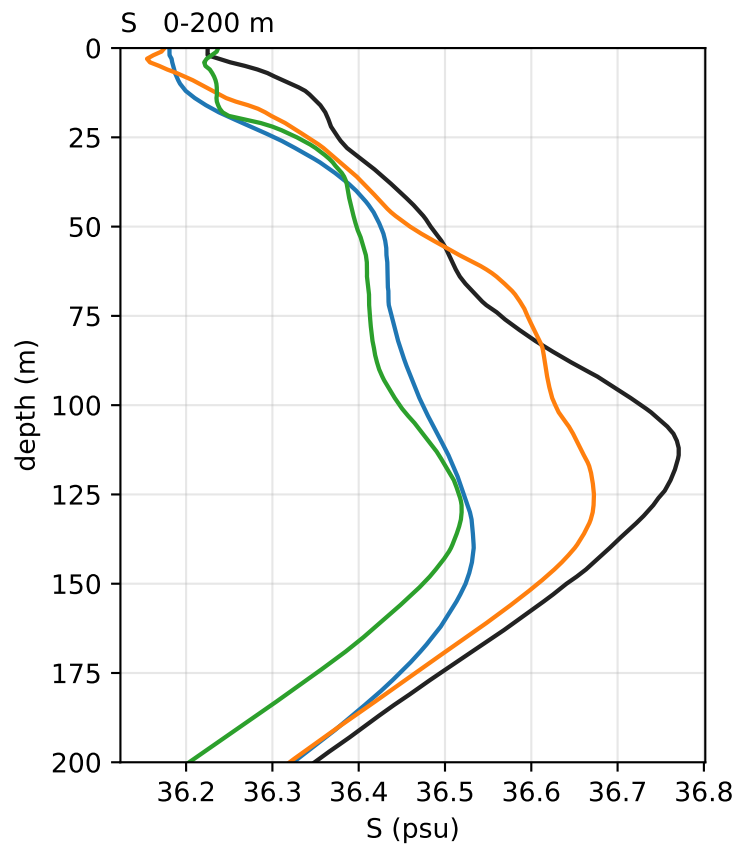
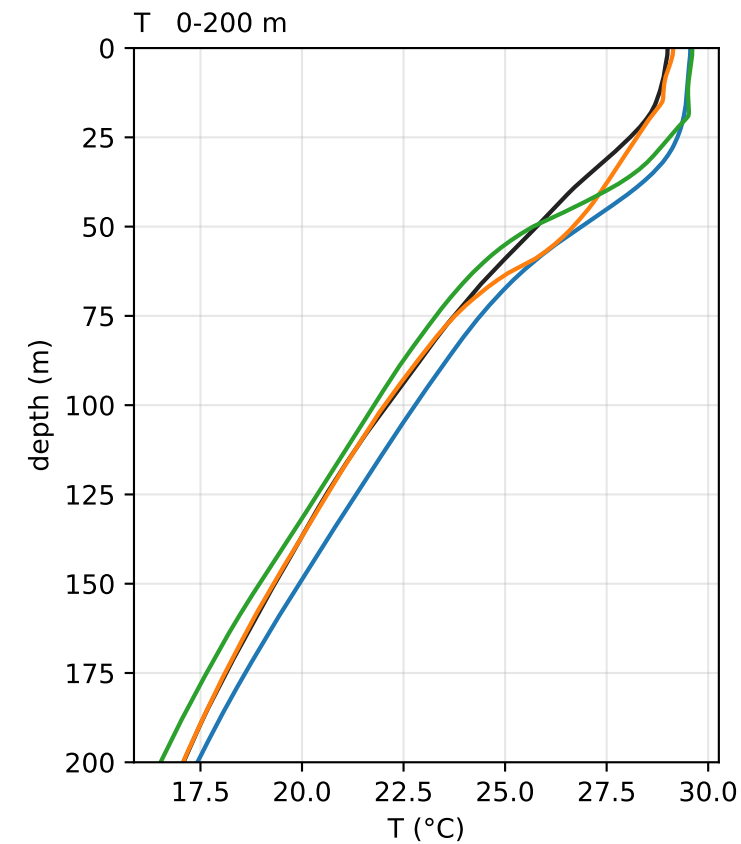
v1 A_CRPS HeaveFast ops

25.00 N, 83.00 W 2025-06-15
east GoM, west of Florida



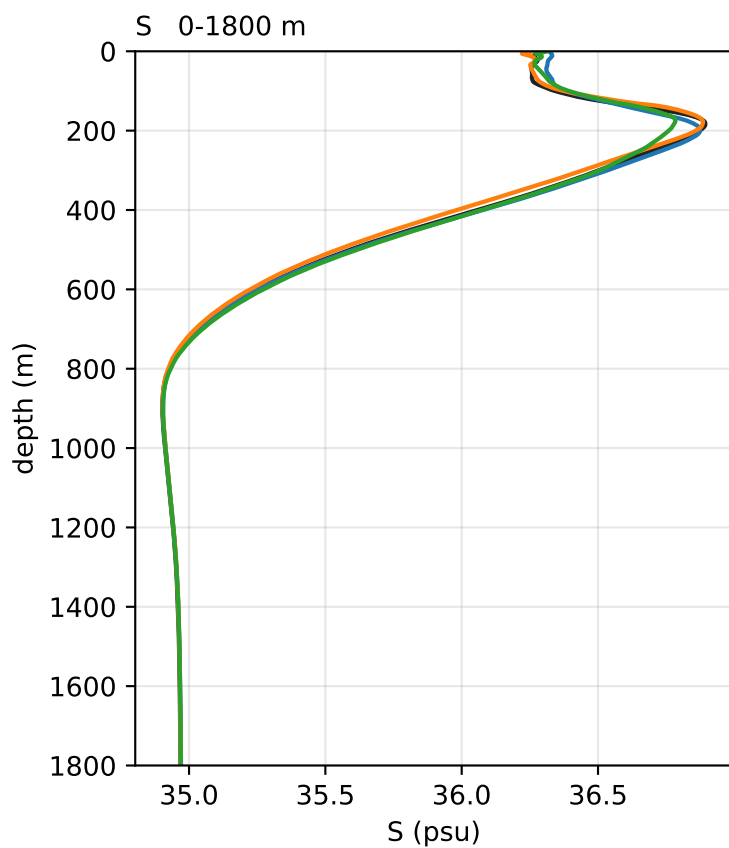
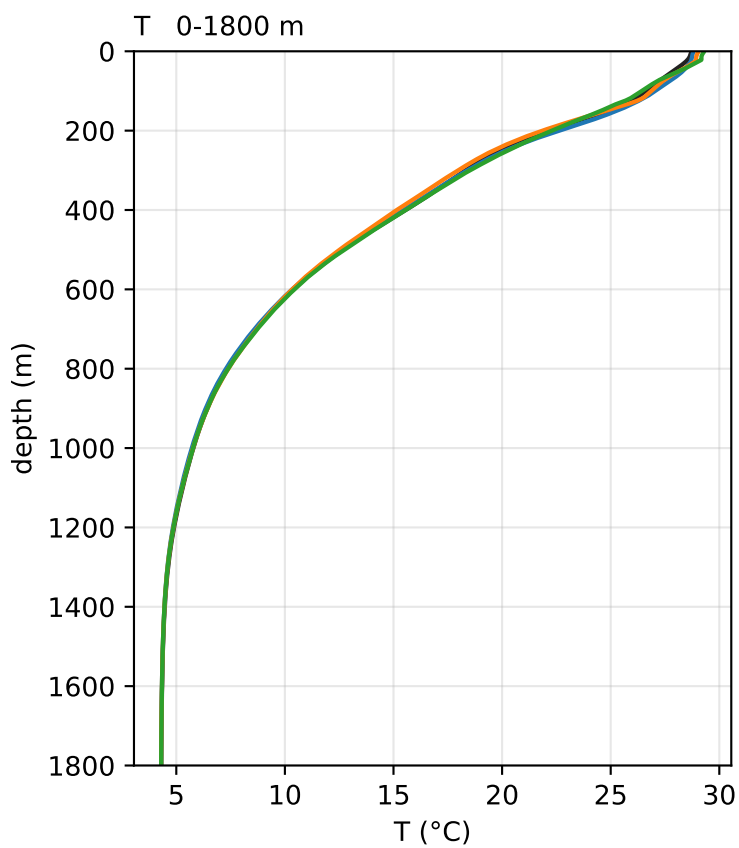
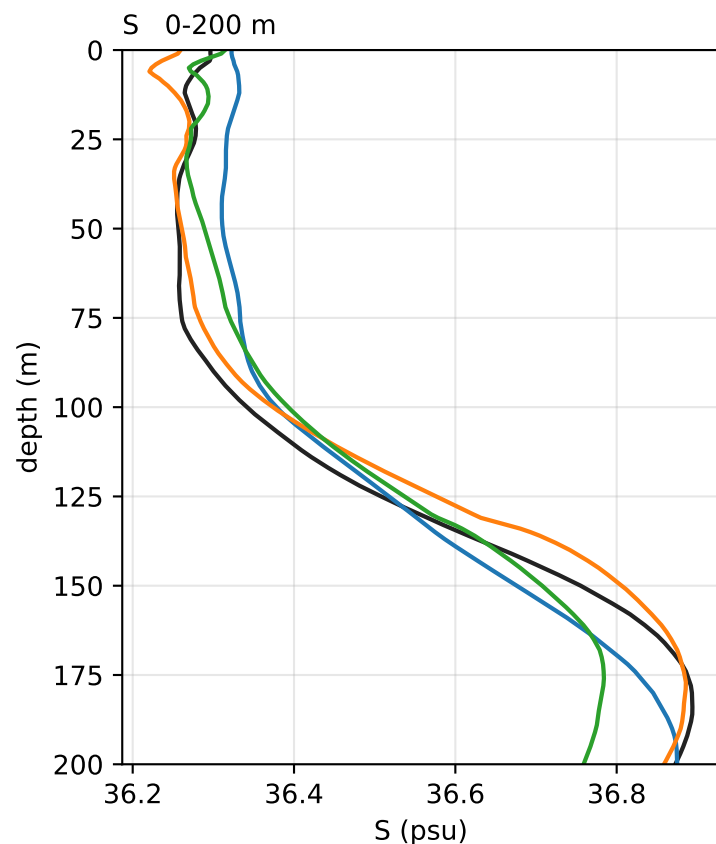
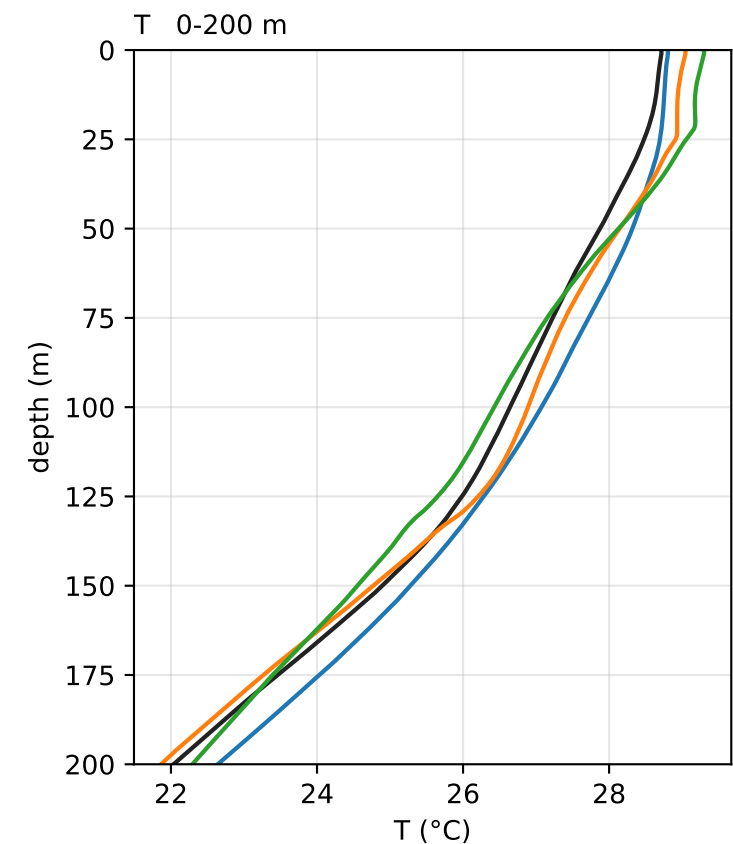
v1 A_CRPS HeaveFast ops

26.50 N, 85.50 W 2025-06-15
north-central GoM



v1 A_CRPS HeaveFast ops

24.80 N, 86.50 W 2025-06-15
south-central GoM



v1 A_CRPS HeaveFast ops

RMSE versus v1 (SAT TorchScript), interpolated onto v1 depth

model	loc	date	T 0-200	S 0-200	T 0-1800	S 0-1800
A_CRPS	25.00N 83.00W	2016-12-15	0.374	0.085	0.179	0.036
A_CRPS	26.50N 85.50W	2016-12-15	0.354	0.067	0.166	0.029
A_CRPS	24.80N 86.50W	2016-12-15	0.466	0.031	0.272	0.029
A_CRPS	25.00N 83.00W	2025-06-15	0.479	0.094	0.179	0.033
A_CRPS	26.50N 85.50W	2025-06-15	0.791	0.143	0.277	0.050
A_CRPS	24.80N 86.50W	2025-06-15	0.424	0.054	0.175	0.021
HeaveFast	25.00N 83.00W	2016-12-15	0.343	0.096	0.128	0.033
HeaveFast	26.50N 85.50W	2016-12-15	0.292	0.055	0.107	0.020
HeaveFast	24.80N 86.50W	2016-12-15	0.106	0.046	0.039	0.016
HeaveFast	25.00N 83.00W	2025-06-15	0.962	0.125	0.368	0.051
HeaveFast	26.50N 85.50W	2025-06-15	0.313	0.058	0.157	0.033
HeaveFast	24.80N 86.50W	2025-06-15	0.242	0.037	0.155	0.029
ops	25.00N 83.00W	2016-12-15	0.415	0.071	0.171	0.029
ops	26.50N 85.50W	2016-12-15	0.265	0.051	0.097	0.020
ops	24.80N 86.50W	2016-12-15	0.317	0.077	0.167	0.031
ops	25.00N 83.00W	2025-06-15	0.386	0.084	0.134	0.030
ops	26.50N 85.50W	2025-06-15	0.534	0.169	0.379	0.079
ops	24.80N 86.50W	2025-06-15	0.329	0.059	0.145	0.026

Units: °C and psu. These are not scores against Argo. They only say how far each v2 cell sits from the production SAT profile at the same point and day. A_CRPS is a 9-d SAT ingest. HeaveFast adds ONI/RONI. ops adds the 19 SAT-archive operators. They should not match v1.